PLC item guide

Units 9 and 10 · the answer to every question on the 12 forms, and what each wrong answer means

At Thursday's PLC

1. Open your school copy of Wednesday's Check: **Responses** → **Summary**. Under each question, Forms shows how many got it right and how many picked each answer.
2. Find the question with the fewest right. Then find the wrong answer most students picked.
3. Look that answer up below: the table says what the mistake is, and the **Reteach** line names the lesson in *03 Reteach lessons*.

Typed answers: Forms marks them; the table lists the wrong values students are most likely to type. Re-check and Review forms list the related lesson for reference.

C-AR.1.1 Check · Simplify linear expressions

5 questions · no calculator · Locked mode on

Key 1 C · 2 A · 3 A · 4 D · 5 D

Question 1

Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$\left(3+\frac{5}{6}p\right)+\left(\frac{5}{6}p-1\right)$$

A. $\frac{5}{6}p+2$

B. $\frac{5}{3}p+4$

C. $\frac{5}{3}p+2$

D. $\frac{11}{3}p$

ANSWER **C** $\frac{5}{3}p+2$

Group like terms: $\frac{5}{6}p+\frac{5}{6}p=\frac{10}{6}p=\frac{5}{3}p$, and $3+(-1)=2$. The expression is $\frac{5}{3}p+2$.

WRONG ANSWERS

A	counted only one of the p-terms
B	treated −1 as +1
D	added the number into the p-term (unlike terms)

Reteach: lesson 1, Adding expressions: match up like terms

Question 2

Which expression is equivalent to the expression shown?

$$(0.4x+2)+(1.6x-2.5)$$

A. $2x-0.5$

B. $1.5x$

C. $0.64x-0.5$

D. $2x+0.5$

ANSWER **A** $2x-0.5$

Add the x-terms: $0.4x+1.6x=2x$. Add the constants: $2-2.5=-0.5$. An x-term and a constant cannot be combined into one term.

WRONG ANSWERS

B	combined the x-terms and the numbers into one term
C	multiplied 0.4×1.6 instead of adding
D	subtracted backwards: $2.5-2=0.5$, but $2-2.5=-0.5$

Reteach: lesson 2, Adding decimal terms: add the numbers in front

Question 3

Which expression makes the equation true?

$$\left(\frac{2}{3}x+6\right)-(__)=-\frac{3}{8}x-2$$

A. $\frac{25}{24}x+8$

B. $\frac{7}{24}x+8$

C. $\frac{25}{24}x+4$

D. $-\frac{25}{24}x-8$

ANSWER **A** $\frac{25}{24}x+8$

The blank is $\left(\frac{2}{3}x+6\right)-\left(-\frac{3}{8}x-2\right)$.

x-terms: $\frac{2}{3}-\left(-\frac{3}{8}\right)=\frac{16}{24}+\frac{9}{24}=\frac{25}{24}$. Constants: $6-(-2)=8$. The expression is $\frac{25}{24}x+8$.

Check: $\left(\frac{2}{3}x+6\right)-\left(\frac{25}{24}x+8\right)=-\frac{9}{24}x-2=-\frac{3}{8}x-2$.

WRONG ANSWERS

B	lost the negative: did $\frac{2}{3}-\frac{3}{8}$ instead of $\frac{2}{3}-\left(-\frac{3}{8}\right)$
C	did $6-2$ instead of $6-(-2)$
D	found right − left, so every sign is flipped

Reteach: lesson 3, The missing expression: undo it with subtraction

---

Question 4

Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$\left(\frac{4}{3}x-8\right)-\left(\frac{7}{3}x+4\right)$$

A. $-x-4$

B. $x-12$

C. $\frac{11}{3}x-4$

D. $-x-12$

ANSWER **D** $-x-12$

Subtracting the second group subtracts both of its terms: $\frac{4}{3}x-8-\frac{7}{3}x-4$.

x-terms: $\frac{4}{3}-\frac{7}{3}=-\frac{3}{3}=-1$, so $-x$. Constants: $-8-4=-12$. The expression is $-x-12$.

WRONG ANSWERS

A	flipped only the x-term of the second group (kept $+4$)
B	subtracted backwards: $\frac{7}{3}-\frac{4}{3}$
C	added the groups instead of subtracting

Reteach: lesson 4, Subtracting a group: the minus reaches every term

---

Question 5 rigor

A rectangle has a length of $\left(\frac{5}{2}x+0.5\right)$ meters and a width of $(x-2)$ meters.

Which expression gives the perimeter of the rectangle, in meters, written with the fewest possible terms?

A. $7x-1.5$

B. $\frac{7}{2}x-1.5$

C. $7x+5$

D. $7x-3$

ANSWER **D** $7x-3$

$\text{Perimeter}=2(\text{length})+2(\text{width})=2\left(\frac{5}{2}x+0.5\right)+2(x-2)$.

Distribute: $5x+1+2x-4$. Combine like terms: $7x-3$.

WRONG ANSWERS

A	doubled the x-terms but not the numbers
B	added length + width once (half the perimeter)
C	added 4 instead of subtracting it

Reteach: lesson 5, From a situation: write it whole, then simplify

C-AR.1.1 Re-check · Simplify linear expressions

3 questions · no calculator · Locked mode on

Key 1 D · 2 D · 3 D

Question 1

Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$(6x - 6) - (0.49x - 5 + 0.58x)$$

A. $4.93x - 11$

B. $6.09x - 11$

C. $7.07x - 1$

D. $4.93x - 1$

ANSWER **D** $4.93x - 1$

Subtracting the group subtracts every term in it: $6x - 6 - 0.49x + 5 - 0.58x$. x-terms: $6 - 0.49 - 0.58 = 4.93$. Constants: $-6 + 5 = -1$. The expression is $4.93x - 1$.

WRONG ANSWERS

A	did not flip the sign of –5
B	flipped only the first term of the group
C	added the x-terms instead of subtracting

Related lesson: 4, Subtracting a group: the minus reaches every term

Question 2

Which expression is equivalent to the expression shown?

$$(3.2x + 2) + (4.5x - 0.5)$$

A. $14.4x + 1.5$

B. $7.7x - 1.5$

C. $9.2x$

D. $7.7x + 1.5$

ANSWER **D** $7.7x + 1.5$

Add the x-terms: $3.2x + 4.5x = 7.7x$. Add the constants: $2 - 0.5 = 1.5$. An x-term and a constant cannot be combined into one term.

WRONG ANSWERS

A	multiplied 3.2×4.5 instead of adding
B	subtracted backwards: $0.5 - 2 = -1.5$, but $2 - 0.5 = 1.5$
C	combined everything into one term

Related lesson: 2, Adding decimal terms: add the numbers in front

Question 3

Which expression makes the equation true?

$$\left(\frac{3}{2}x + 8\right) - (__) = -\frac{5}{3}x - 1$$

A. $-\frac{19}{6}x - 9$

B. $\frac{19}{6}x + 7$

C. $-\frac{1}{6}x + 9$

D. $\frac{19}{6}x + 9$

ANSWER **D** $\frac{19}{6}x + 9$

The blank is $\left(\frac{3}{2}x + 8\right) - \left(-\frac{5}{3}x - 1\right)$.

x-terms: $\frac{3}{2} - \left(-\frac{5}{3}\right) = \frac{9}{6} + \frac{10}{6} = \frac{19}{6}$. Constants: $8 - (-1) = 9$. The expression is $\frac{19}{6}x + 9$.

Check: $\left(\frac{3}{2}x + 8\right) - \left(\frac{19}{6}x + 9\right) = -\frac{10}{6}x - 1 = -\frac{5}{3}x - 1$.

WRONG ANSWERS

A	found right – left (every sign flipped)
B	did $8 - 1$ instead of $8 - (-1)$
C	lost the negative: $\frac{3}{2} - \frac{5}{3}$ instead of $\frac{3}{2} - \left(-\frac{5}{3}\right)$

Related lesson: 3, The missing expression: undo it with subtraction

C-AR.1.1 Review · Simplify linear expressions

2 questions · no calculator · Locked mode on

Key 1 D · 2 C

Question 1

Which expression is equivalent to the expression shown, written with the fewest possible terms?

$$(2x - 4) - (0.59x - 1 + 0.44x)$$

A. $3.03x - 3$

B. $0.97x - 5$

C. $1.85x - 5$

D. $0.97x - 3$

ANSWER **D** $0.97x - 3$

Subtract every term in the group: $2x - 4 - 0.59x + 1 - 0.44x$.
x-terms: $2 - 0.59 - 0.44 = 0.97$. Constants: $-4 + 1 = -3$. The expression is $0.97x - 3$.

WRONG ANSWERS

A	did not flip the signs of the x-terms in the group
B	did not flip the sign of –1
C	flipped only the first term of the group

Related lesson: 4, Subtracting a group: the minus reaches every term

Question 2

Which expression makes the equation true?

$$\left(\frac{5}{3}x - 7\right) + (__) = \frac{7}{4}x - 2$$

A. $\frac{1}{12}x - 9$

B. $-\frac{1}{12}x - 5$

C. $\frac{1}{12}x + 5$

D. $\frac{41}{12}x - 9$

ANSWER **C** $\frac{1}{12}x + 5$

The blank is $\left(\frac{7}{4}x - 2\right) - \left(\frac{5}{3}x - 7\right)$.
x-terms: $\frac{7}{4} - \frac{5}{3} = \frac{21}{12} - \frac{20}{12} = \frac{1}{12}$. Constants: $-2 - (-7) = 5$. The expression is $\frac{1}{12}x + 5$.
Check: $\left(\frac{5}{3}x - 7\right) + \left(\frac{1}{12}x + 5\right) = \frac{21}{12}x - 2 = \frac{7}{4}x - 2$.

WRONG ANSWERS

A	did $-2 + (-7)$ instead of $-2 - (-7)$
B	found left – right (every sign flipped)
D	added the expressions instead of subtracting

Related lesson: 3, The missing expression: undo it with subtraction

C-AR.1.2 Check · Determine if two expressions are equivalent

5 questions · no calculator · Locked mode on

Key 1 A+C · 2 D · 3 B · 4 D · 5 B

Question 1

Select **all** the expressions that are equivalent to $\frac{5}{4}(12x + 20)$.

*Select **all** the correct answers.*

A. $\frac{5}{4}(12x) + \frac{5}{4}(20)$

B. $\frac{53}{4}x + \frac{85}{4}$

C. $15x + 25$

D. $15x + 20$

E. $40x$

ANSWER A and C

Checkbox question: one point only for exactly A and C.

Distribute $\frac{5}{4}$ to both terms: $\frac{5}{4} \times 12x = 15x$ and $\frac{5}{4} \times 20 = 25$.
So $\frac{5}{4}(12x + 20) = 15x + 25$, which is also written $\frac{5}{4}(12x) + \frac{5}{4}(20)$.

WRONG ANSWERS

B	added $\frac{5}{4}$ to each term instead of multiplying
D	multiplied only the first term
E	combined $15x$ and 25 (unlike terms)

Reteach: lesson 1, Distributing a fraction: multiply every term

Question 2

A student rewrote $\frac{4}{3}(12x + 12)$ as $28x$.

The correct equivalent expression is ____.

Choose the expression that correctly completes the sentence.

A. $16x + 12$

B. $28x$

C. $\frac{40}{3}x + \frac{40}{3}$

D. $16x + 16$

ANSWER D $16x + 16$

The student multiplied only $12x$ by $\frac{4}{3}$, which gives $16x + 12$, and then added $16x$ and 12 as if they were like terms.
Both terms must be multiplied by $\frac{4}{3}$, and an x-term and a constant cannot be combined: $\frac{4}{3}(12x + 12) = 16x + 16$.

WRONG ANSWERS

A	multiplied only the first term
B	repeated the student's error
C	added $\frac{4}{3}$ to each term instead of multiplying

Reteach: lesson 2, Spot the error: multiply both terms, then stop

Question 3

Which expression is equivalent to $1.5(6x - 2)$?

A. $9x - 2$

B. $9x - 3$

C. $6x$

D. $7.5x - 0.5$

ANSWER B $9x - 3$

Distribute 1.5 to both terms inside the parentheses.
$1.5 \times 6x = 9x$ and $1.5 \times (-2) = -3$, so the result is $9x - 3$.

WRONG ANSWERS

A	multiplied only the first term
C	combined $9x$ and –3 (unlike terms)
D	added 1.5 instead of multiplying

Reteach: lesson 3, Distributing a decimal: same rule, careful multiplying

Question 4

Are the two expressions equivalent?

$2.5(8x - 2) + 4x + 8$ and $24x + 3.5$

A. Yes, because both of them simplify to $24x + 3.5$.

B. No, because $2.5(8x - 2)$ is $20x - 2$, giving $24x + 6$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.

ANSWER D No, because the first simplifies to $24x + 3$, not $24x + 3.5$.

Distribute and combine: $2.5(8x - 2) = 20x - 5$, then + $4x + 8$ gives $24x + 3$.
The second expression is $24x + 3.5$. The constants differ (3 and 3.5), so they are not equivalent.
The 2.5 multiplies both terms in the parentheses.

WRONG ANSWERS

A	said yes without simplifying
B	right answer (no) for a wrong reason: multiplied only $8x$ by 2.5
C	judged by looks: same form is not the same value

Reteach: lesson 4, Equivalent or not: simplify both, then compare

Question 5 rigor

Two expressions are shown.

$-\frac{3}{4}(4x-4)+\frac{3}{2}x$ and $\frac{11}{2}x-\frac{3}{4}(8x-4)$

Are the two expressions equivalent? Choose the answer with the correct reason.

A. Yes; both have a constant term of 3.

B. No; the first simplifies to $-\frac{3}{2}x+3$ and the second to $-\frac{1}{2}x+3$.

C. Yes; both simplify to $-\frac{3}{2}x+3$.

D. No; the first simplifies to $-\frac{3}{2}x-3$ and the second to $-\frac{1}{2}x+3$.

ANSWER **B** No; the first simplifies to $-\frac{3}{2}x+3$ and the second to $-\frac{1}{2}x+3$.

First: $-\frac{3}{4}(4x-4)+\frac{3}{2}x=-3x+3+\frac{3}{2}x=-\frac{3}{2}x+3$.
Second: $\frac{11}{2}x-\frac{3}{4}(8x-4)=\frac{11}{2}x-6x+3=-\frac{1}{2}x+3$.
The x-terms are different, so the expressions are not equivalent. Having the same constant term is not enough, and $-\left(\frac{3}{4}\right)\times(-4)$ is +3, not −3.

WRONG ANSWERS

A	the same number at the end is not enough
C	simplified only the first and assumed the second matched; it is $-\frac{1}{2}x+3$
D	in the first expression, took $-\frac{3}{4}(-4)$ as -3 instead of $+3$

Reteach: lesson 5, A negative outside the parentheses: flip every sign inside

C-AR.1.2 Re-check · Determine if two expressions are equivalent

3 questions · no calculator · Locked mode on

Key 1 D · 2 A · 3 A+E

Question 1

Are the two expressions equivalent?

$1.5(4x-6)+x+8$ and $7x-2$

A. No, because $1.5(4x-6)$ is $6x-6$, giving $7x+2$.

B. Yes, because both of them simplify to $7x-2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x-1$, not $7x-2$.

ANSWER **D** No, because the first simplifies to $7x-1$, not $7x-2$.

Distribute and combine: $1.5(4x-6)=6x-9$, then + $x+8$ gives $7x-1$.
The second expression is $7x-2$. The constants differ $(-1 \text{ and } -2)$, so they are not equivalent.
The 1.5 multiplies both terms in the parentheses.

WRONG ANSWERS

A	multiplied only $4x$ by 1.5
B	said yes without simplifying
C	judged by looks: same form is not the same value

Related lesson: 4, Equivalent or not: simplify both, then compare

Question 2

Which expression is equivalent to $6x+7.5$?

A. $1.5(4x+5)$

B. $6(x+5)$

C. $1.5(6x+5)$

D. $1.5(4x+7.5)$

ANSWER **A** $1.5(4x+5)$

Both terms of $6x+7.5$ are 1.5 times a whole number: $6=1.5\times4$ and $7.5=1.5\times5$.
So $6x+7.5=1.5(4x+5)$. Check by distributing: $1.5\times4x=6x$ and $1.5\times5=7.5$.

WRONG ANSWERS

B	took a different number out of each term: 6 from $6x$, but 1.5 from 7.5
C	left 6 inside instead of $6\div1.5=4$
D	divided only the first term by 1.5

Related lesson: 4, Equivalent or not: simplify both, then compare

Question 3

Select **all** the expressions that are equivalent to $\frac{3}{2}(6x+2)$.

*Select **all** the correct answers.*

A. $\frac{3}{2}(6x)+\frac{3}{2}(2)$

B. $\frac{15}{2}x+\frac{7}{2}$

C. $12x$

D. $9x+2$

E. $9x+3$

ANSWER **A and E**

Checkbox question: one point only for exactly A and E.

Distribute $\frac{3}{2}$ to both terms: $\frac{3}{2}\times 6x=9x$ and $\frac{3}{2}\times 2=3$.
So $\frac{3}{2}(6x+2)=9x+3$, which is also written $\frac{3}{2}(6x)+\frac{3}{2}(2)$.

WRONG ANSWERS

B	added $\frac{3}{2}$ to each term instead of multiplying
C	combined $9x$ and 3 (unlike terms)
D	multiplied only the first term

Related lesson: 1, Distributing a fraction: multiply every term

C-AR.1.2 Review · Determine if two expressions are equivalent

2 questions · no calculator · Locked mode on

Key 1 B · 2 A

Question 1

Which expression is equivalent to $8x+6$?

A. $\frac{2}{5}(20x+6)$

B. $\frac{2}{5}(20x+15)$

C. $\frac{5}{2}(20x+15)$

D. $\frac{2}{5}(8x+15)$

ANSWER **B** $\frac{2}{5}(20x+15)$

Both terms of $8x+6$ are $\frac{2}{5}$ times a whole number: $8=\frac{2}{5}\times 20$ and $6=\frac{2}{5}\times 15$.
So $8x+6=\frac{2}{5}(20x+15)$. Check by distributing: $\frac{2}{5}\times 20x=8x$ and $\frac{2}{5}\times 15=6$.

WRONG ANSWERS

A	divided only the first term by $\frac{2}{5}$
C	used the reciprocal, $\frac{5}{2}$
D	left 8 inside instead of $8\div\left(\frac{2}{5}\right)=20$

Related lesson: 4, Equivalent or not: simplify both, then compare

Question 2

A student rewrote $\frac{7}{4}(12x+4)$ as $25x$.

The correct equivalent expression is ____.

Choose the expression that correctly completes the sentence.

A. $21x+7$

B. $\frac{55}{4}x+\frac{23}{4}$

C. $21x+4$

D. $25x$

ANSWER **A** $21x+7$

The student multiplied only $12x$ by $\frac{7}{4}$, which gives $21x+4$, and then added $21x$ and 4 as if they were like terms.
Both terms must be multiplied by $\frac{7}{4}$, and an x-term and a constant cannot be combined: $\frac{7}{4}(12x+4)=21x+7$.

WRONG ANSWERS

B	added $\frac{7}{4}$ to each term instead of multiplying
C	multiplied only the first term
D	repeated the student's error

Related lesson: 2, Spot the error: multiply both terms, then stop

C-AR.2.2 Check · Two-step equations with rational numbers

5 questions · calculator allowed · Locked mode off

Key 1 D · 2 −2 · 3 A · 4 3 1/3 · 5 B

Question 1

A park charges a $25.00 entry fee plus $12.50 for each hour of skate rental. A visitor paid $62.50 in all.

Which choice shows an equation that can be used to find h, the number of hours of skate rental the visitor paid for, and its solution?

A. $12.5h - 25 = 62.5;\ h = 7$

B. $25h + 12.5 = 62.5;\ h = 2$

C. $12.5h + 25 = 62.5;\ h = 7$

D. $12.5h + 25 = 62.5;\ h = 3$

ANSWER **D** $12.5h + 25 = 62.5;\ h = 3$

Total = entry fee + $12.50 for each hour: $12.5h + 25 = 62.5$.
Subtract 25: $12.5h = 37.5$. Divide by 12.5: $h = 3$.
Check: $12.5 \times 3 + 25 = 62.5$.

WRONG ANSWERS

A	subtracted the entry fee in the equation
B	swapped the hourly rate and the fee
C	right equation, but added 25 instead of subtracting

Reteach: lesson 1, Fee plus rate: write it, then undo in reverse order

---

Question 2

Solve the equation for h.

$$1.5(h + 1.6) = -0.6$$

Type a number only. No units, $ or %.

ANSWER $h = -2$

The form accepts −2

Divide both sides by 1.5: $h + 1.6 = -0.6 \div 1.5 = -0.4$.
Subtract 1.6 from both sides: $h = -0.4 - 1.6 = -2$.

WRONG ANSWERS STUDENTS TYPE

1.2	added 1.6 at the end instead of subtracting
−2.5	multiplied by 1.5 instead of dividing
−2.2	subtracted 1.6 but never divided by 1.5
−1.47 or $-\frac{22}{15}$	multiplied only h by 1.5

Reteach: lesson 2, Equations with parentheses: undo the outside first

---

Question 3

Three students each saved the same amount, s dollars, and then each earned $5.00 more. Together they now have $105.00.

Which choice shows an equation that can be used to find s, and its solution?

A. $3(s + 5) = 105;\ s = 30$

B. $3s + 5 = 105;\ s = 33\frac{1}{3}$

C. $3(s + 5) = 105;\ s = 35$

D. $3(s - 5) = 105;\ s = 40$

ANSWER **A** $3(s + 5) = 105;\ s = 30$

Each student now has $s + 5$ dollars, and 3 students together have $105: $3(s + 5) = 105$.
Divide by 3: $s + 5 = 35$. Subtract 5: $s = 30$.
Check: $3 \times (30 + 5) = 105$.

WRONG ANSWERS

B	added the $5 once instead of for each student
C	stopped at $s + 5 = 35$
D	subtracted the $5 instead of adding

Reteach: lesson 3, Groups: when the change happens to each one

Question 4

What is the exact value of g in the equation shown?

$$-1\frac{7}{9} = \frac{2}{3}g - 4$$

Type a number only. No units, $ or %.

ANSWER $g = 3\frac{1}{3}$

The form accepts 3 1/3 or 10/3

Add 4 to both sides: $-1\frac{7}{9} + 4 = 2\frac{2}{9}$, so $2\frac{2}{9} = \frac{2}{3}g$.
Divide both sides by $\frac{2}{3}$, which means multiplying by its reciprocal $\frac{3}{2}$: $g = \frac{20}{9} \times \frac{3}{2} = 3\frac{1}{3}$.

WRONG ANSWERS STUDENTS TYPE

$5\frac{2}{3}$ **or** $\frac{17}{3}$	read $-1\frac{7}{9}$ as $-1 + \frac{7}{9}$
$1\frac{13}{27}$ **or** $\frac{40}{27}$	multiplied by $\frac{2}{3}$ instead of $\frac{3}{2}$
$-8\frac{2}{3}$ **or** $-\frac{26}{3}$	subtracted 4 instead of adding
$2\frac{2}{9}$ **or** $\frac{20}{9}$	stopped before dividing by $\frac{2}{3}$

Reteach: lesson 4, Fraction equations: add, then use the reciprocal

Question 5 rigor

Gym A charges $24.50 each month. Jordan used a coupon worth $35.00 off the total and paid Gym A $185.50 in all.

Let n be the number of months Jordan paid for. Gym B charges $26.50 each month and takes no coupons. Which choice gives n, the gym that would have cost less for the same number of months, and how much less?

A. $n = 6$; Gym B, by $26.50

B. $n = 9$; Gym A, by $53.00

C. $n = 9$; Gym A, by $18.00

D. $n = 8$; Gym A, by $26.50

ANSWER **B** $n = 9$; Gym A, by $53.00

The months cost $24.50 \times n$, and the coupon takes off $35.00: $24.5n - 35 = 185.5$.
Add 35: $24.5n = 220.5$. Divide by 24.5: $n = 9$.
Gym B for 9 months: $26.50 \times 9 = \$238.50$, with no coupon. Jordan paid Gym A $185.50, so Gym A cost less, by $238.50 - 185.50 = \$53.00$.

WRONG ANSWERS

A	added the coupon instead of subtracting $(24.5n + 35)$
C	compared monthly prices and left out the coupon
D	divided 185.50 by 24.50 and rounded

Reteach: lesson 5, Two-part problems: solve, then answer the real question

C-AR.2.2 Re-check · Two-step equations with rational numbers

3 questions · calculator allowed · Locked mode off

Key 1 –9 · 2 A · 3 A

Question 1

Solve the equation for x.

$$40 = -4x + 4$$

Type a number only. No units, $ or %.

ANSWER $x = -9$

The form accepts –9

Subtract 4 from both sides: $36 = -4x$.
Divide both sides by –4: $x = -9$.

WRONG ANSWERS STUDENTS TYPE

9	dropped the negative: $36 \div (-4)$ is –9
–11	added 4 instead of subtracting
–10	divided 40 by –4 and ignored the + 4

Related lesson: 1, Fee plus rate: write it, then undo in reverse order

Question 2

Six identical boxes each weigh b pounds with the packaging. After 1.25 pounds of packaging is removed from each box, the 6 boxes weigh 40.5 pounds in all.

Which choice shows an equation that can be used to find b, and its solution?

A. $6(b - 1.25) = 40.5;\ b = 8$

B. $b - 1.25 = 40.5;\ b = 41.75$

C. $6(b - 1.25) = 40.5;\ b = 6.75$

D. $6(b + 1.25) = 40.5;\ b = 5.5$

ANSWER **A** $6(b - 1.25) = 40.5;\ b = 8$

Each box weighs $b - 1.25$ pounds after the packaging is removed, and 6 boxes weigh 40.5 pounds: $6(b - 1.25) = 40.5$.
Divide by 6: $b - 1.25 = 6.75$. Add 1.25: $b = 8$.
Check: $6 \times (8 - 1.25) = 6 \times 6.75 = 40.5$.

WRONG ANSWERS

B	forgot there are 6 boxes
C	stopped at $b - 1.25 = 6.75$
D	added the packaging instead of taking it away

Related lesson: 3, Groups: when the change happens to each one

Question 3

A number, x, is decreased by 8. That result is then multiplied by 3, giving 3.

Which equation represents this situation?

A. $3(x - 8) = 3$

B. $3x - 24 = 9$

C. $3(x + 8) = 3$

D. $3x - 8 = 3$

ANSWER **A** $3(x - 8) = 3$

"Decreased by 8" is $(x - 8)$.
Multiplying that result by 3 gives $3(x - 8)$, and that equals 3.

WRONG ANSWERS

B	distributed correctly, then multiplied the right side by 3 too $(3 \times 3 = 9)$
C	read "decreased" as +
D	multiplied only x by 3

Related lesson: 3, Groups: when the change happens to each one

C-AR.2.2 Review · Two-step equations with rational numbers

2 questions · calculator allowed · Locked mode off

Key 1 A · 2 2 1/3

Question 1

A park charges a $15.00 entry fee plus $3.75 for each game. A visitor paid $33.75 in all.

Which choice shows an equation that can be used to find g, the number of games the visitor paid for, and its solution?

A. $3.75g + 15 = 33.75;\ g = 5$

B. $15g + 3.75 = 33.75;\ g = 2$

C. $3.75g + 15 = 33.75;\ g = 13$

D. $3.75g - 15 = 33.75;\ g = 13$

ANSWER **A** $3.75g + 15 = 33.75;\ g = 5$

Total = entry fee + $3.75 for each game: $3.75g + 15 = 33.75$.
Subtract 15: $3.75g = 18.75$. Divide by 3.75: $g = 5$.
Check: $3.75 \times 5 + 15 = 33.75$.

WRONG ANSWERS

B	swapped the price per game and the fee
C	added 15 instead of subtracting
D	subtracted the fee in the equation

Related lesson: 1, Fee plus rate: write it, then undo in reverse order

Question 2

What is the exact value of t in the equation shown?

$$-5\frac{8}{9} = \frac{4}{3}t - 9$$

Type a number only. No units, $ or %.

ANSWER $t = 2\frac{1}{3}$

The form accepts 2 1/3 or 7/3

Add 9 to both sides: $-5\frac{8}{9} + 9 = 3\frac{1}{9}$, so $3\frac{1}{9} = \frac{4}{3}t$.

Divide both sides by $\frac{4}{3}$, which means multiplying by its reciprocal $\frac{3}{4}$: $t = \frac{28}{9} \times \frac{3}{4} = 2\frac{1}{3}$.

WRONG ANSWERS STUDENTS TYPE

$3\frac{2}{3}$ **or** $\frac{11}{3}$	read $-5\frac{8}{9}$ as $-5 + \frac{8}{9}$
$4\frac{4}{27}$ **or** $\frac{112}{27}$	multiplied by $\frac{4}{3}$ instead of $\frac{3}{4}$
$3\frac{1}{9}$ **or** $\frac{28}{9}$	stopped before dividing by $\frac{4}{3}$
$-11\frac{1}{6}$ **or** $-\frac{67}{6}$	subtracted 9 instead of adding

Related lesson: 4, Fraction equations: add, then use the reciprocal

C-AR.2.1 Check · Solve one-step inequalities

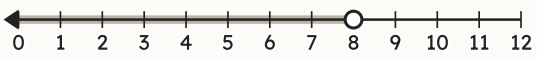
5 questions · calculator allowed · Locked mode off

Key 1 D · 2 D · 3 A · 4 A · 5 D

Question 1

The graph shows the solution of an inequality.

Which inequality has this solution?



A. $2.5x \leq 20$

B. $x + 2.5 < 20$

C. $2.5x > 20$

D. $2.5x < 20$

ANSWER **D** $2.5x < 20$

The open circle at 8 shows that 8 is not included, and the shading to the left shows $x < 8$.
$2.5x < 20$ has this solution: dividing both sides by 2.5 gives $x < 8$.

WRONG ANSWERS

A	closed circle $(\leq)$ for an open circle
B	did not solve it: treated $x + 2.5$ like $2.5x$; it gives $x < 17.5$
C	wrong direction

Reteach: lesson 1, Reading a graph: the circle and the arrow

Question 2

Solve the inequality.

$$-3 + t \geq -6.9$$

What is the solution?

A. $t \geq 3.9$

B. $t \leq -3.9$

C. $t \geq -9.9$

D. $t \geq -3.9$

ANSWER **D** $t \geq -3.9$

Add 3 to both sides: $t \geq -6.9 + 3$, so $t \geq -3.9$.
Adding the same number to both sides does not reverse the inequality symbol.

WRONG ANSWERS

A	dropped the negative: $-6.9 + 3$ is −3.9
B	flipped the symbol after only adding
C	subtracted 3 instead of adding it

Reteach: lesson 2, Adding or subtracting: the symbol stays put

Question 3

Solve the inequality.

$$-0.5n > 0.5$$

Which number line shows the solution?

A. Open circle at −1, shaded left

B. Closed circle at −1, shaded left

C. Open circle at 1, shaded left

D. Open circle at −1, shaded right

ANSWER **A** Open circle at −1, shaded left

Divide both sides by −0.5. Dividing by a negative number reverses the inequality symbol, so $n < -1$.
An open circle at −1 shows that −1 is not included, and the shading goes to the left.

WRONG ANSWERS

B	closed circle for a strict $>$
C	dropped the negative: $0.5 \div (-0.5)$ is −1
D	did not flip after dividing by a negative

Reteach: lesson 3, Dividing by a negative flips the symbol

Question 4

The product of a number x and −0.5 is less than −5.

What are the possible values of x?

A. $x > 10$

B. $x > 2.5$

C. $x < -10$

D. $x < 10$

ANSWER **A** $x > 10$

The product of x and −0.5 is less than −5: $-0.5x < -5$.
Divide both sides by −0.5. Dividing by a negative number reverses the inequality symbol: $x > 10$.
Check: $x = 12$ gives $-0.5 \times 12 = -6$, which is less than −5.

WRONG ANSWERS

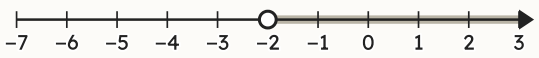
B	multiplied −5 by −0.5 instead of dividing
C	ignored the negative in $-0.5x$: divided by 0.5, so −10 and no flip
D	did not flip after dividing by a negative

Reteach: lesson 4, Words to an inequality: product, sum, at least, at most

Question 5 rigor

The number line shows the solution of one of the inequalities below.

Which inequality has this solution?



A. $x + 3.5 \geq 1.5$

B. $\frac{x}{4} < -0.5$

C. $x - 4.5 > -8.5$

D. $-1.5x < 3$

ANSWER **D** $-1.5x < 3$

The graph shows $x > -2$: an open circle at −2, shaded to the right.
Solve each choice. Only $-1.5x < 3$ gives $x > -2$ (dividing by the negative coefficient reverses the symbol).
The other three solve to a different direction, the other kind of circle, or a different boundary.

WRONG ANSWERS

A	gives $x \geq -2$: closed circle
B	flipped because −0.5 is negative; $\times 4$ (positive) keeps $<$, so it gives $x < -2$
C	gives $x > -4$: wrong boundary

Reteach: lesson 5, Matching a graph: solve every choice, check three things

C-AR.2.1 Re-check · Solve one-step inequalities

3 questions · calculator allowed · Locked mode off

Key 1 A · 2 A · 3 D

Question 1

The product of a number x and −1.5 is less than −12.

What are the possible values of x?

A. $x > 8$

B. $x > 18$

C. $x < -8$

D. $x < 8$

ANSWER **A** $x > 8$

The product of x and −1.5 is less than −12: $-1.5x < -12$.
Divide both sides by −1.5 and reverse the symbol: $x > 8$.
Check: $x = 10$ gives $-1.5 \times 10 = -15$, which is less than −12.

WRONG ANSWERS

B	multiplied −12 by −1.5 instead of dividing
C	ignored the negative in $-1.5x$: divided by 1.5, so −8 and no flip
D	did not flip after dividing by a negative

Related lesson: 4, Words to an inequality: product, sum, at least, at most

Question 2

Solve the inequality.

$$-1.5c \leq 4.5$$

Which number line shows the solution?

A. Closed circle at −3, shaded right

B. Closed circle at 3, shaded right

C. Open circle at −3, shaded right

D. Closed circle at −3, shaded left

ANSWER **A** Closed circle at −3, shaded right

Divide both sides by −1.5. Dividing by a negative number reverses the inequality symbol, so $c \geq -3$.
A closed circle at −3 shows that −3 is included, and the shading goes to the right.

WRONG ANSWERS

B	dropped the negative: $4.5 \div (-1.5)$ is −3
C	open circle for $\geq$
D	did not flip after dividing by a negative

Related lesson: 3, Dividing by a negative flips the symbol

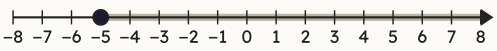
Question 3

Solve the inequality.

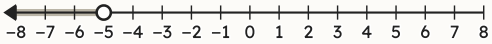
$$\frac{n}{-4} \geq 1.25$$

Which number line shows the solution?

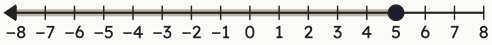
A



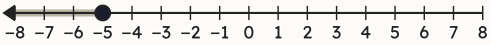
B



C



D



ANSWER **D**

Multiply both sides by −4. Multiplying by a negative number reverses the inequality symbol: $n \leq 1.25 \times (-4)$, so $n \leq -5$.
−5 is included $(\leq)$, so the circle at −5 is closed, and the solutions less than −5 are shaded to the left: number line D.

WRONG ANSWERS

A	did not flip after multiplying by −4
B	open circle for $\leq$
C	dropped the negative: $1.25 \times (-4)$ is −5

Related lesson: 3, Dividing by a negative flips the symbol

C-AR.2.1 Review · Solve one-step inequalities

2 questions · calculator allowed · Locked mode off

Key 1 C · 2 C

Question 1

The product of a number x and −2.5 is at least −20.

What are the possible values of x?

A. $x \leq 50$

B. $x \geq 8$

C. $x \leq 8$

D. $x < 8$

ANSWER **C** $x \leq 8$

"At least −20" means greater than or equal to −20: $-2.5x \geq -20$.
Divide both sides by −2.5 and reverse the symbol: $x \leq 8$.
Check: $x = 8$ gives −20, which is at least −20, and $x = 4$ gives −10, which is also at least −20.

WRONG ANSWERS

A	multiplied −20 by −2.5 instead of dividing
B	did not flip
D	strict $<$ instead of $\leq$ ("at least" includes 8)

Related lesson: 4, Words to an inequality: product, sum, at least, at most

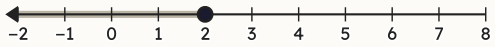
Question 2

Solve the inequality.

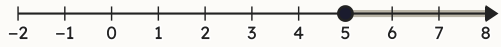
$$0.5w \leq 2.5$$

Which number line shows the solution?

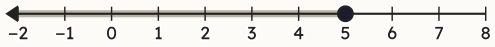
A



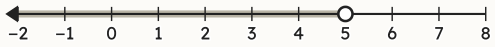
B



C



D



ANSWER **C**

Divide both sides by 0.5: $w \leq 2.5 \div 0.5$, so $w \leq 5$. Dividing by a positive number does not reverse the symbol.
5 is included $(\leq)$, so the circle at 5 is closed, and the solutions less than 5 are shaded to the left: number line C.

WRONG ANSWERS

A	subtracted 0.5 instead of dividing
B	flipped when dividing by a positive
D	open circle for $\leq$

Related lesson: 3, Dividing by a negative flips the symbol